Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2005
PI Name:	Butel, Janet S. Ph.D.		
Project Title:	Biology of Virus Infections: Radiation and Immunity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030-3411	**Congressional District:**	25
Comments:	Address updated 9/2008		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Changed end date of3/31/2008.		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Conner, Margaret (Baylor College of Medicine) Ling, Paul (Baylor College of Medicine) Zhang, Shaojie (Baylor College of Medicine)		
Grant/Contract No.:	NCC 9-58-IIH00403		
Performance Goal No.:			
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Task Description:	Space flight has been found to affect immune responses, although the underlying mechanisms have not been established. Alterations in a normal immune response could have a major impact on the host's ability to control infections. An important question being explored is whether infectious diseases will pose a medical risk to the success of long-duration space journeys. All humans are infected for life with latent and persistent viruses, and it is well-known that suppression of the immune system allows latent viruses to reactivate and multiply, which may cause disease in the person undergoing reactivation or in contacts to whom the virus is transmitted. The general hypothesis being addressed is that conditions of long-duration space flight, including stress, isolation, containment, microbial contamination, and solar radiation, will alter human immune responses, leading to reactivation of latent viruses, increased viral infections and viral disease, and possible development of malignancies. We are focusing on reactivation and shedding of human herpesvirus EBV and human polyomaviruses, agents known to establish persistent infections and to undergo reactivation and cause disease, including cancer, when the host immune system is compromised. Animal models are being used to study radiation effects on viral infections and host responses to those infections. The specific aims of the project are: (1) To determine the effects of space radiation and hind limb unloading (HLU) on host control of virus infections and virus-induced cancers; and (2) To characterize direct effects of radiation on viruses and virus-infected cells. Key findings of the project this year were from the mouse polyoma virus space radiation model we developed. A quantitative real-time polymerase chain reaction assay was developed to measure murine polyoma virus genome copies in infected animal tissues. Preliminary studies suggest that combined effects of radiation and virus infection on experimental animals lead to immunosuppression and enhanced virus replication. Another key finding was the identification of a herpesvirus EBV latent protein that might modulate the development of EBV-associated disease. These findings may allow new approaches, such as novel drugs, to prevent and treat virus reactivation and virus-associated diseases that arise against an immunosuppressed host background. Plans for the coming year include experiments to determine the effect of multiple, low-dose radiation exposures on host control of viral infections; measurements of virus reactivations in volunteers participating in the NASA-sponsored Bed Rest Study, a model for the effects of space flight conditions on human physiology, in collaboration with Dr. G. Sonnenfeld; studies of radiation-induced signals that cause EBV reactivation from latency in cultured cells; tests of whether the effects of HLU and irradiation synergistically affect control of polyoma virus infections (reactivation) and immune impairment; and determination by flow cytometry of whether virus infection, irradiation, and HLU of mice results in altered distribution and numbers of individual leukocyte cell subsets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The mouse model we developed here involving polyoma virus infection and radiation will allow targeted studies of the effects of irradiation on host immune function, virus infection, and tumor development. Such data will define the risk of these combined factors to long-duration space flight and will allow tests of countermeasures. We have shown that radiation can cause reactivation of latent viral infections and depress host immune function. The evidence for viral reactivations highlights the need to understand synergistic effects of radiation and space flight conditions that may result in more deleterious effects on immune function and control of microbial infections and disease. The knowledge gained from studies of virus reactivations in these test models will be applicable to earth-bound individuals at risk of suffering similar virus reactivations and serious, sometimes life-threatening, consequences due to immunosuppression following organ transplantation or cancer chemotherapy and during pregnancy, old age, and AIDS. Studies of the role of EBV latent genes in human disease may lead to the development of novel antiviral drugs.
Task Progress:	Key findings of the project this year were from the mouse polyoma virus space radiation model we developed. A quantitative real-time polymerase chain reaction assay was developed to measure murine polyoma virus genome copies in infected animal tissues. Preliminary studies suggest that combined effects of radiation and virus infection on experimental animals lead to immunosuppression and enhanced virus replication. Another key finding was the identification of a herpesvirus EBV latent protein that might modulate the development of EBV-associated disease. These findings may allow approaches, such as novel drugs, to prevent and treat virus reactivation and virus-associated diseases that arise against an immunosuppressed host background.
Bibliography Type:	Description: (Last Updated: 04/25/2011)
Articles in Peer-reviewed Journals	Forsman ZH, Lednicky JA, Fox GE, Willson RC, White ZS, Halvorson SJ, Wong C, Lewis AM Jr, Butel JS. "Phylogenetic analysis of polyomavirus simian virus 40 from monkeys and humans." J Virol. 2004 Sep;78(17):9306-16. PMID: 15308725 , Sep-2004
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Articles in Peer-reviewed Journals	Shearer WT, Zhang S, Reuben JM, Lee BN, Butel JS. "Effects of radiation and latent virus on immune responses in a space flight model." J Allergy Clin Immunol. 2005 Jun;115(6):1297-303. PMID: 15940150 , Jun-2005
Articles in Peer-reviewed Journals	Walling DM, Ling PD, Gordadze AV, Montes-Walters M, Flaitz CM, Nichols CM. "Expression of Epstein-Barr virus latent genes in oral epithelium: determinants of the pathogenesis of oral hairy leukoplakia." J Infect Dis. 2004 Jul 15;190(2):396-9. Epub 2004 Jun 11. PMID: 15216478 , Jul-2004